

DR. SUDESH PAHAL

HEAD & ASSOCIATE PROFESSOR | DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Maharaja Surajmal Institute of Technology (GGSIIP University), C-4, Janakpuri, New Delhi | **Phone: +91-8588964280** |
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PROFESSIONAL PROFILE

Academic and research professional with 20+ years of teaching, academic administration, innovation mentoring and research experience in Electronics & Communication Engineering. Current research interests include wireless and vehicular networks, machine learning, deep learning, digital signal processing and data analytics, with publications across SCI/SCIE, Scopus and other peer-reviewed journals. Experienced in institutional innovation, accreditation, student mentoring, industry-linked training, hackathons, research guidance and academic programme development.

PROFESSIONAL EXPERIENCE

Head, Department of Electronics & Communication Engineering, Maharaja Surajmal Institute of Technology, New Delhi | June 2026–Present

Associate Professor, Maharaja Surajmal Institute of Technology, New Delhi | August 2016–Present

Assistant Professor, Maharaja Surajmal Institute of Technology, New Delhi | August 2011–August 2016

Lecturer, Guru Premsukh Memorial College of Engineering, New Delhi | January 2009–August 2011

Lecturer, Bhagwan Mahaveer Institute of Technology, Sonapat | January 2008–January 2009

Lecturer, Bhagwan Mahaveer Institute of Technology, Sonapat | August 2005–November 2006

ACADEMIC QUALIFICATIONS

Ph.D. "Cross Layer Trigger Based Handover Schemes in Wireless Networks" Manav Rachna International University, Faridabad | 2014

M.E. (ECE) Deenbandhu Chhoturam University of Science and Technology, Murthal | 2008

B.E. (ECE) Bhagwan Mahaveer Institute of Technology, Sonapat | 2005

HSC CBSE | 2001

SSC HBSE | 1999

AREAS OF EXPERTISE

Analog & Digital Communications • Digital Signal Processing • Data Analytics • Machine Learning • Deep Learning • Vehicular Networks • Wireless Networks

AWARDS, HONORS & RECOGNITION

- Conducted 200-hour Wipro certification training (June–September 2026) for 73 students.
- Project "ASTRA-Low-Cost IoT-Enabled Antenna Solution for Women's Security Devices" selected for funding and mentoring under IPU-IIF (June 2026).
- Served as Session Chair at ICESS 2026, 2–3 March 2026, the 1st International Electronics Packaging Design & Manufacturing Conference.
- Received a Letter of Appreciation for serving as Coordinator for the FDP "Developing Healthy Work Culture at Workplace", 2–6 December 2024, conducted by NITTTR Chandigarh.
- Project "Portable ICU" selected for mentoring as an incubation mentor/partner at IPU-IIF (June 2025).
- Editorial Board Member, SCI-indexed journal "Scientific Reports", Springer Nature (September 2025).
- Project "Saarthi" selected for mentoring as an incubation mentor/partner at IPU-IIF (June 2025).
- Recognized as Wipro Certified Faculty for Project-Based Learning (PBL) in Data Science with Python after clearing the TalentNext certification assessment with 87.5% (December 2025).
- Honoured by Dr. Kiran Bedi with a certificate and ₹11,000 for securing Second Runner-Up in the National Innovation Award (November 2024).
- Received First Prize and ₹11,000 in the inter-college competition "Ignite & Innovate", BMIET Sonapat (2025).

- Best Paper Award, NCI-TIDE 2025.
- Silver Certificate, NPTEL course on "Deep Learning" (2024).
- Elite Certificate, NPTEL course on "Data Analysis using Python" (2025).
- Academic Excellent Performance Certificate for outstanding performance in academics and administration at MSIT (2023).
- Elite Certificate, NPTEL course on "Data Science using Python" (2023).
- Research Excellence Certificate in recognition of outstanding contribution to research quality (2022, 2023).
- Best Paper Award, ADMET 2021.
- Certified Innovation Ambassador and mentor of nationally selected start-ups by MHRD (2020).
- Faculty Representative Member, Governing Body, Surajmal Memorial Education Society (2017–2019).
- Certificate of Appreciation from the Chairman, Surajmal Memorial Education Society, for outstanding contribution to certificates awarded to MSIT by the Government of India (2022).
- Mentor of Smart India Hackathon winner team (2023); National Innovation Challenge finalist (2020, 2023); SIH finalist (2024, 2025).
- Certificate of Appreciation from MoE Innovation Cell for achieving 3.5 stars for IIC 2023–24 and 2024–25.
- Merit/Distinction certificates/degrees in HSC, B.E. and M.E. (1999, 2005 and 2008, respectively).

RESEARCH, OUTREACH & COLLABORATION

- Lifetime Member, ISTE Technical Society; Innovation Ambassador, MHRD Innovation Cell.
- Received ₹50,000 grant under the Unnat Bharat Abhiyan scheme of MHRD (2018).
- Organizing Member, AICTE/ISTE-sponsored refresher programme "Skill Development for Entrepreneurs" (2022), with grant of ₹83,000.
- Recognized as Co-Supervisor of one Ph.D. scholar at Manav Rachna International University, Faridabad.
- Delivered invited lectures on Intelligent Transportation Systems, innovation and entrepreneurship, Mentor–Mentee Orientation and related themes at University of Delhi and Govt. SPMR College, Jammu (2021–2024).
- Session Coordinator at ADMET 2021, NCETET 2024, ICAIA 2025 and NCI-TIDE 2025.
- Coordinator for linkage between MSIT and the ATL of Andhra Education Society NTRM Senior Secondary School, Janakpuri (2023).
- Mentor Coordinator, Bhagini Nivedita College, Delhi (2024).
- Received ₹54,000 grant from MoE, Government of India, for hosting Junior SIH as a nodal centre.
- Organized a special session at ICICC 2025; served as Session Chair at ICICC 2025, NCI-TIDE 2025 and ICESS 2026.
- Mentor of a finalist team at the national-level HACKNATION 2026, Shivalik College, Dehradun.

SELECTED PUBLICATIONS – SCI/SCIE & SCOPUS JOURNALS

1. Pahal, S., & Nandal, P. (2025). "A Hybrid Framework for Kinematic Vehicle Prediction: YOLOv11 with Unscented Kalman Filter." *SAE International Journal of Connected and Automated Vehicles*.
2. S. Pahal, Saloni, Umesh (2025). "A Review of Different Strategies for Vehicle Collision Avoidance." *International Journal of Vehicle Information and Communication Systems*, 10(3), 299–324. DOI: 10.1504/IJVICIS.2025.147512.
3. Nandal, P., Pahal, S. & Upadhyay, G.M. (2025). "Image Denoising Using Quantum Deep Convolutional Generative Adversarial Network for Medical Images." *International Journal of Computational Intelligence Systems*, 18, 190. DOI: 10.1007/s44196-025-00920-6.
4. Nandal, P., Pahal, S., Sharma, T., & Omesh, O. (2025). "Real-Time Driver Drowsiness Detection Using YOLOv8 with Whale Optimization Algorithm." *SAE International Journal of Transportation Safety*, 13(1), 15–33. DOI: 10.4271/09-13-01-0001.
5. Nandal, P., Pahal, S., Malik, S. et al. (2025). "Enhancing real time object detection for autonomous driving using YOLO-NAS algorithm with CLEO optimizer." *International Journal of Information Technology*, 17, 1321–1328. DOI: 10.1007/s41870-024-02296-w.
6. Nandal, P., Pahal, S., Khanna, A., & Pinheiro, P.R. (2024). "Super-Resolution of Medical Images Using Real ESRGAN." *IEEE Access*, 12, 176155–176170. DOI: 10.1109/ACCESS.2024.3497002.
7. Pahal, S., Nandal, P., Gupta, N., & Sehrawat, N. (2024). "Multiple Criteria-Based Intelligent Techniques for Efficient Handover in Next-Generation Networks." *International Journal of Sensors, Wireless Communications and Control*.
8. Nandal, P., Pahal, S., Gupta, S., Khanna, A., & Ali, T. (2025). "Lightweight ESRGAN-Based Underwater Image Quality Enhancement." *Journal of Instrumentation*, 20. DOI: 10.1088/1748-0221/20/08/P08009.

9. Gupta, N., Shekhar, C., Pahal, S., Baghel, S.K., & Nandal, P. (2024). "Realtime Driver Drowsiness Detection using YOLOv5 and Computer Vision Techniques." *Journal of Computational Analysis and Applications*.
10. Gupta, N., Jagriti, Shuchi Aggarwal, Pahal, S., & Jain, R. (2024). "Digitalization in Operations Management Practices in Organizations." *Journal of Computational Analysis and Applications*.
11. Siwach, M., Nandal, P., & Pahal, S. (2025). "Coindonate: A blockchain-based platform for transparent online donation system." *Journal of Information & Optimization Sciences*, 46(1), 157–166. DOI: 10.47974/JIOS-1860.
12. Pahal, S., Rathee, N., & Singh, B. (2023). "A Deep Learning-Based Model for Link Quality Estimation in Vehicular Networks." *IETE Journal of Research*, 69(8), 5159–5168. DOI: 10.1080/03772063.2021.1973591.
13. Dhochak, M., Pahal, S., & Doliya, P. (2022). "Predicting the Startup Valuation: A deep learning approach." *Venture Capital*. DOI: 10.1080/13691066.2022.2161968.
14. Rathee, N., Pahal, S., & Sheoran, D. (2022). "Pain detection from facial expressions using domain adaptation technique." *Pattern Analysis and Applications*, 25(3), 567–574. DOI: 10.1007/s10044-021-01025-4.
15. Pahal, S., & Rathee, N. (2021). "Evaluating uncertainty of measurement while predicting location in smart vehicles." *MAPAN – Journal of Metrology Society of India*, 36, 377–388. DOI: 10.1007/s12647-021-00458-w.
16. Rathee, N., Pahal, S., & Sheoran, D. (2021). "Evaluating the Uncertainty of Classification Caused Due to Image Resizing Techniques for Satellite Image Classification." *MAPAN – Journal of Metrology Society of India*. DOI: 10.1007/s12647-021-00456-y.
17. Pahal, S. (2017). "Effect of Velocity on Handover Parameters in Wireless Networks." *International Journal of Systems, Control and Communications*, 8, 177–186. DOI: 10.1504/IJSCC.2017.083994.
18. Pahal, S., Singh, B., & Arora, A. (2015). "Cross Layer based Dynamic Handover Decision in Heterogeneous Wireless Networks." *Wireless Personal Communications*, 82, 1665–1684. DOI: 10.1007/s11277-015-2305-6.
19. Pahal, S., Singh, B., & Arora, A. (2014). "Cross layer based Fast Handover Algorithm for IEEE 802.16e Networks." *Optik*, 125, 4108–4112. DOI: 10.1016/j.ijleo.2014.01.104.
20. Pahal, S., Singh, B., & Arora, A. (2015). "Cross layer based Handover Mechanism in Mobile WiMAX Networks." *International Journal of Ad-hoc and Ubiquitous Computing*, 19(3), 133–142. DOI: 10.1504/IJAHUC.2015.070588.
21. Pahal, S., Singh, B., & Arora, A. (2013). "A Predictive Handover Initiation Mechanism in Next Generation Wireless Networks." *WSEAS Transactions on Communications*, 12(6), 271–281.

UGC / REFEREED JOURNAL PUBLICATIONS

- Pahal, S. & Anjana (2023). "Distributed Denial of Services attacks on cloud servers: Detection, Analysis and Mitigation." *Mapana Journal of Sciences*, 22(1), 121–145. DOI: 10.12723/mjs.64.7.
- Pahal, S. & Preity P. (2022). "A Comprehensive Research Study on Low-Interaction Secure Shell HoneyPot." *Mapana Journal of Sciences*, 21(4), 99–118.
- Pahal, S. & Rathee, N. (2016). "Analysis of Hysteresis Margin for Effective Handover in 4G Wireless Networks." *ICTACT Journal on Communication Technology*, 7(4), 1438–1442.
- Pahal, S., Singh, B., & Arora, A. (2012). "A Novel Approach for Handover Trigger Time Selection in WiMAX Networks." *International Journal of Advanced Electronics & Communication Systems*, 2(2).
- Pahal, S., Singh, B., & Arora, A. (2014). "Performance Evaluation of Vertical Handover based on Signal Strength and Residual Time in Heterogeneous Wireless Networks." *International Journal of Computing and Network Technology*, 2(1), 25–31.
- Pahal, S. & Sehrawat, P. (2015). "Multiple Attribute Based Handoff Network Selection Techniques In Heterogeneous Networks." *IOSR Journal of Mobile Computing and Application*, 2(2), 46–55.
- Pahal, S., Agnihotri, A., & Mendiratta, D. (2019–20). "Identity Management & Access Authorization using Multi-dimensional Block-Chain." *Satyam-MSIT Journal of Research*, 8, 8–10.
- Pahal, S. et al. (2019–20). "Nanotechnology for Powerful Solar Energy." *Satyam-MSIT Journal of Research*, 4, 8–10.

CONFERENCE PAPERS / PROCEEDINGS

- Rohilla, V., Pahal, S., & Nandal, P. (2025). "Deep LocAWS: A Deep Multilayer Perceptron Neural Network Collaborative Filtering for Location-Aware Web Service Recommendation." *ICICC 2025, Springer LNNS*, Vol. 1432. DOI: 10.1007/978-981-96-6715-4_28.
- Pahal, S., Sehrawat, N., Joshi, V., Pranshu, & Mishra, P.K. (2025). "Developing a Machine Learning Framework for Optimizing Electric Vehicle Charging Infrastructure Deployment in India." *ICICC 2025, Springer LNNS*, Vol. 1439. DOI: 10.1007/978-981-96-7137-3_8.

- Pahal, S., Prasad, A., Khokhar, N., & Gupta, N. (2025). "Enhancing Lung Cancer Diagnosis Through CNN-Based Analysis of Clinical and Histopathological Data." ICICC 2025, Springer LNNS, Vol. 1432. DOI: 10.1007/978-981-96-6715-4_4.
- Saloni, U. Dutta, & S. Pahal (2025). "Evaluating the Assessment of CNN and YOLO Based Models for Detecting Road Accidents." ICDISS 2025. DOI: 10.1109/ICDISS68238.2025.11320762.
- Pahal, S. & Nandal, P. (2024). "Handover Challenges in Wireless Communications and Their Solutions." ICICC 2024, SSRN.
- Pahal, S., Singh, B., & Arora, A. (2013). "A Prediction Based Handover Trigger in Overlapped Heterogeneous Wireless Networks." IEEE ISPPC 2013, JUIT, Shimla.
- Pahal, S. & Singh, B. "Cross Layer Trigger based Mobility Management in Wireless Networks." International Conference on Advancements in Communication, Computing and Signal Processing, Gautam Buddha University.
- Pahal, S., Pandove, G., & Kumar, A. (2008). "A WiMAX module with security features using NS2." WECON-08.
- Pahal, S., Pandove, G., & Kumar, A. (2008). "A review on WiMAX Security." NCETET-08.
- Pahal, S., Pandove, G., & Kumar, A. (2008). "An improved WiMAX Security Scheme based on Asymmetric key approach." IEEE AIS-2008.
- Pahal, S., Gulia, V., Chauhan, N., & Goel, A. (2016). "Analysis of Handoff factors in Wireless Networks." IEEE ICCCA-2016, pp. 631–636.
- Pahal, S., Singh, B., & Arora, A. (2013). "QoS and Velocity based Vertical Handover Decision Algorithm using Fuzzy Logic for Mobile Wireless Networks."
- Pahal, S., Deswal, M., & Gupta, A. (2017). "Implementation of prediction algorithms in MIP handover." ICFESMH 2017.
- Pahal, S. (2018). "Design and Analysis of Double Spurline Bandstop Filters." ICETEST-2018.
- Pahal, S., Vaishali, & Tripti (2019). "Vehicular Communication using Radio Waves: A step towards Smarter vehicles." RTCAIT 2019.
- Pahal, S. (2019). "Miniaturized Microstrip Bandstop Filters with Wider Bandwidth." ICAEECI 2019.
- Pahal, S. et al. (2024). "Decision making based on Business analysis using R language." ICTII 2024.
- Pahal, S. (2025) "Evaluating the Assessment of CNN and YOLO based models for detecting road accidents", IEEE conf. ICDISS-2025 at MRIU Faridabad, 14-15 Nov,2025.
- Pahal, S. (2025), "Performance Measurement and Accuracy Evaluation of Integrated Multi-Band Antennas on a Single PCB for IoT Devices", NCI-TIDE 2025, 18-19 Nov 2025.

BOOKS & BOOK CHAPTERS

- Pahal, S. (2013). Cross Layer Based Handover Trigger Generation in Wireless Networks. Lambert Academic Publishing, Germany. ISBN: 978-3-659-43951-3.
- Pahal, S. (2015). Handover Analysis for Different Mobile Propagation Models. Lambert Academic Publishing, Germany. ISBN: 978-3-659-85253-4.
- Pahal, S. et al. (2018). "Securing Onion Routing against Correlation attack." Advances in Intelligent Systems and Computing, Springer, Vol. 698. ISBN: 978-9-811-31818-4.
- Nandal, P. & Pahal, S. (2024). "Upscaling images using ESRGAN with perceptual loss." ICICC 2024, Springer LNNS, Web of Science.
- Pahal, S., Rathee, N. (2020). "Investigation of facial expressions for physiological parameter measurement." Advances in Computing and Intelligent Systems, Springer, pp. 297–302.
- Pahal, S. & Sehrawat, P. (2020). "Image Colorization with Deep Convolutional Neural Networks." Advances in Communication and Computational Technology, Springer, pp. 45–56. ISBN: 978-981-15-5341-7.

PATENTS

- "Point to point regression analysis-based big data processing using machine learning." Patent ID: 202111017897 | Date: 21 May 2021.
- "Computer Based Irish Detection Device for Biometric Authentication." Design No. 409905-001 | Date: 09 March 2024.
- "Accident Prevention System for Vehicles Using Deep Learning." Patent ID: 202411037276 | Date: 31 May 2024.

ADMINISTRATIVE RESPONSIBILITIES & INSTITUTIONAL CONTRIBUTIONS

- Convener, Institute Innovation Council (IIC), MSIT (2018–Present). Established the IIC in 2018; IIC MSIT received 3, 4, 3.5 and 3-star ratings from the Ministry of Education, Government of India, for 2019–2022.

- Presented IIC poster at the IIC Regional Meet, Amity University, Noida (2023).
- IIC MSIT linked with Atal Tinkering Labs of Andhra Education Society NTRM Senior Secondary School and Veda Vyasa DAV Public School.
- Selected as Mentor and Member of the IIC State Consortium by MoE, Government of India (2023).
- Conducted hackathons and mentored teams that won Smart India Hackathon and reached the final stage of the National Innovation Challenge (2019–Present).
- Eight start-ups established as a member of the formulation and implementation team of the National Innovation and Startup Policy & Entrepreneurship Cell (2020–Present).
- More than 40 innovative student ideas/prototypes registered on the YUKTI portal; nine teams selected for demonstration at the YUKTI Innovation Challenge.
- Two teams selected for the national bootcamp organized by MoE, Government of India, at IIT Guwahati (2022).
- One team won ₹40,000 in the national hackathon conducted by Yamaha.
- Eight faculty members trained as Innovation Ambassadors.
- One team won the International Indo-Africa Hackathon with a prize of ₹3 lakh.
- Coordinator, National Intellectual Property Awareness Mission (NIPAM), MSIT (2022); received certificate of appreciation from the Ministry of Commerce and Industry, Government of India.
- Convener, Unnat Bharat Abhiyan, MSIT (2018–2021): established the UBA cell, received ₹50,000 grant from MHRD, adopted five villages, and conducted surveys, awareness programmes and technical projects.
- Criteria Coordinator, NBA (2018–Present): established mentoring process and Student Support System contributing to NBA accreditation.
- Lab In-charge (2018–Present): introduced new experiments on DSO, spectrum analyser and Python; upgraded and maintained the Communication Systems Laboratory.
- Convener, Tech Fest (2017): successfully organized AVENSIS 2017 at MSIT.
- Academic Programme Committee Member (2021–Present): contributed to the new scheme for ECE students; Subject Coordinator for DSP and Mobile Communication, GGSIP University.
- In-charge, Quiz Society “Vedas”, MSIT (2020–Present): organized inter- and intra-college competitions.
- NIRF Coordinator & In-charge, Institute Branding Committee (2018–2020).
- Coordinator: organized more than 40 FDPs, industrial visits, workshops and seminars for students and faculty; attended more than 50 FDPs and workshops.
- ISO Internal Auditor: 2022–23, 2023–24, 2024–25 and 2025–26.
- E-Summit Co-Convener.
- NBA Criteria 9 (Student Support System) Mentoring In-charge (2018–Present).
- NBA Coordinator (Aug 2025 onwards)

PROFESSIONAL DEVELOPMENT

- NITTTR FDP: “Accomplishment of Curriculum Objectives using FOSS”, 9–13 March 2026 (one week).
- Microsoft Power BI Data Analyst Associate 2025 (1 week)
- NPTEL course Data Analytics with Python 2025 (12 weeks)
- FDP attended Developing Healthy Work Culture at Work Place 2025 (1 week)
- Course on Data Science with python by Wipro 2025 (12 weeks)

DECLARATION

I hereby declare that the information furnished above is true and complete to the best of my knowledge and belief.

Place: Delhi | Date: 25 August 2026

Dr. Sudesh Pahal